

# When Senses Collide: The Science of Synesthesia and Why Some People See Music

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Imagine biting into a lemon and suddenly seeing a flash of purple. Or hearing a violin note and tasting strawberries. For most of us, that sounds like fantasy. But for people with **synesthesia**, it's everyday reality. **Synesthesia** is a fascinating **neurological** trait in which stimulation of one sensory pathway triggers an involuntary experience in another. It's not a disorder; rather, it's considered a form of **neurodiversity**, affecting up to 4 percent of the population. The most common form is **grapheme-color synesthesia**, where letters or numbers are perceived as inherently colored. For instance, the letter 'A' might always appear red, or the number '7' consistently yellow. About 1 percent of people have this specific type. Another variant, **chromesthesia**, links sound to color or shape. Historians suspect Vincent van Gogh had this trait; he described seeing colors when he played piano keys. Contemporary musician Billie Eilish has also spoken about her synesthetic experiences, telling iHeart Radio, "Everything that I make I'm already thinking of what color it is, and what texture it is, and what day of the week it is, and what number it is, and what shape." Scientists believe synesthesia has a genetic component, often running in families. Brain imaging studies suggest it may result from increased cross-wiring or reduced inhibition between sensory regions. While the exact mechanisms are still debated, one thing is clear: synesthesia offers a unique window into how the brain constructs reality. It reminds us that perception is deeply personal and that the world can be a much more colorful, flavorful, and musical place for some.

## Word Treasure

**synesthesia** -- A brain condition where stimulating one sense automatically triggers an experience in another sense.

**neurological** -- Related to the brain and nervous system.

**neurodiversity** -- The idea that differences in brain function, like synesthesia or autism, are natural variations and not necessarily disorders.

**grapheme-color synesthesia** -- A type of synesthesia where letters or numbers are perceived as having specific colors.

**chromesthesia** -- A form of synesthesia in which sounds, especially music, trigger the perception of colors or shapes.

**cross-wiring** -- A simplified explanation for synesthesia meaning extra connections or less inhibition between different brain areas.

## Background you need

Synesthesia was first documented scientifically in the late 1800s by Francis Galton, but for much of history it was often dismissed as imagination or hallucination.

In the 20th century, cognitive scientists confirmed it as a genuine neurological phenomenon through tests showing that synesthetes' associations are consistent over time.

Famous confirmed or suspected synesthetes include composer Franz Liszt, who saw colors when composing, and author Vladimir Nabokov, who wrote about letters having specific hues.

Billie Eilish (a Grammy-winning singer born in 2001) has publicly described her vivid synesthetic perceptions, helping bring modern attention to the trait.

Brain imaging studies using fMRI and diffusion tensor imaging have revealed structural differences in the brains of synesthetes, supporting the idea of extra connectivity between sensory areas.

## Break it down

**LEAD:** vivid, relatable examples -- biting a lemon and seeing purple, hearing a violin and tasting strawberries

+ hook: 'For most of us, that sounds like fantasy. But for people with synesthesia, it's everyday reality.'

**DEFINITION:** synesthesia as a neurological trait, not a disorder, and a form of neurodiversity

+ STATISTIC: affects up to 4% of people

### COMMON TYPES

+ GRAPHIC COLOR: letters/numbers always appear in specific colors (~1% of people)

+ CHROMESTHESIA: sound triggers colors/shapes

### REAL-WORLD EXAMPLES

+ HISTORICAL: Vincent van Gogh reportedly saw colors when playing piano

+ CONTEMPORARY: Billie Eilish thinks of music in terms of color, texture, day, and number

### SCIENTIFIC INVESTIGATION

+ GENETIC LINK: runs in families

+ BRAIN MECHANISM: likely increased cross-wiring or reduced inhibition between sensory regions

**CLOSING REFLECTION:** synesthesia reveals how personal perception is -- 'the world can be a much more colorful, flavorful, and musical place for some'

## Why it matters

Synesthesia challenges the assumption that everyone perceives the world in the same way, encouraging empathy and acceptance of neurodiversity. Understanding it could unlock clues about creativity, brain plasticity, and how sensory information is integrated--all crucial as we design future technologies that interface directly with the brain.

## Test what you remember

### 1. What is synesthesia?

- ☐ (A) A disorder that harms one sense
- ☐ (B) A trait where stimulation of one sense triggers another
- ☐ (C) A type of memory loss
- ☐ (D) A sleep disorder

### 2. According to the article, what is the most common form of synesthesia?

- ☐ (A) Chromesthesia
- ☐ (B) Auditory-tactile synesthesia
- ☐ (C) Grapheme-color synesthesia
- ☐ (D) Mirror-touch synesthesia

### 3. Up to what percentage of the population is affected by synesthesia, based on the text?

- ☐ (A) 1 percent
- ☐ (B) 4 percent
- ☐ (C) 10 percent
- ☐ (D) 0.1 percent

### 4. Which famous artist does the article suspect had chromesthesia?

- ☐ (A) Pablo Picasso
- ☐ (B) Claude Monet
- ☐ (C) Vincent van Gogh
- ☐ (D) Leonardo da Vinci

### 5. What does Billie Eilish think about in relation to her music, as quoted?

- ☐ (A) Only the volume and tempo
- ☐ (B) What color, texture, day of the week, and number it is
- ☐ (C) The price of the instruments
- ☐ (D) The weather when she records

### 6. What do scientists believe might cause synesthesia in the brain?

- ☐ (A) A lack of sleep
- ☐ (B) Increased cross-wiring or reduced inhibition between sensory regions
- ☐ (C) Eating too much chocolate
- ☐ (D) Damage from loud music

Answer key (try the quiz first!): 1: B 2: C 3: B 4: C 5: B 6: B

## Questions to think about

**1. Positive / Creative Advantage:** Many synesthetes report that their extra perceptions enrich art, memory, and problem-solving. Musicians like Billie Eilish and historical figures like van Gogh used their sensory blending to create works that resonate deeply with others.

2. **Negative / Overstimulation:** For some, synesthesia can be overwhelming--uncontrollable extra sensations may cause distraction, sensory fatigue, or anxiety in intense environments like busy streets or loud concerts. Society often fails to recognize such invisible challenges.

**3. Neutral / Scientific:** From a research standpoint, synesthesia offers a window into how the brain wires itself. Studying it helps neuroscientists map normal sensory processing, learn about neural plasticity, and possibly develop therapies for conditions where sensory integration goes wrong.

**4. Forward?Looking: As brain?computer interfaces and virtual reality become more sophisticated, insights from synesthesia could guide the creation of multisensory experiences that feel natural. Companies might one day design devices that let users 'see' sounds or 'taste' colors to convey information, especially for people with disabilities.**

**MY THOUGHTS (at least 20 words):**